

Department of Electrical and Computer Engineering
University of Wisconsin–Madison

ECE 553: Testing and Testable Design of Digital Systems
Spring 2025

Project - Test Generation for
Full and Partial Scan

Date: April 8
Phase 1 Due Date: April 15
Phase 2 Due Date: April 29
Project Weight: 20%

Overview:

Our project is composed of two parts: (1) test pattern generation for full scan, and (2) design of partial scan. A report is submitted for each part in addition to some supporting files. For the first part of the project, you will be working alone (one person team). In the second part, there is option to work in teams of three.

The project grade is 20% of your final grade. The grade of the first part is 5% of the and the second part is 15%.

(1) Test Pattern Generation for Full Scan:

You are required to develop a test set for a given circuit (b14.vhd) in the full scan mode. This is a viper processor from the ITC'99 benchmark suite. The design file may be copied from: `cp ~adavoodi/setup/b14.vhd .`

Make sure you carefully examine clock and reset signal names in this design (and other details, e.g., if reset is active high or active low) and make sure these are correctly incorporated to generate the synthesis commands for the full scan mode.

When generating the test set, assume all stuck faults in TetraMax (`add_faults -all`). The test developed must be economical in terms of number of patterns but need not be absolutely minimum in size (see if you can apply test compression when running ATPG). Create a tcl script for use with TetraMax by copying the commands printed in the TetraMax prompt window after each action. You can refer to the TetraMax user manual posted on the Canvas project directory to figure how to do test compression.

In this project we measure the test quality with “Test Coverage” metric given in the ATPG report. Just FYI, we could have also asked ATPG to report “Fault Coverage” (`set_faults -fault_coverage`). TetraMax calculates Test Coverage as below:

$$\#Detected/(\#Total_Faults-\#Undetectable)$$

You are encouraged to explore different strategies, based on the input arguments of the ATPG tool. Examples include controlling the abort limit, use of random test patterns before deterministic testing, use of compaction techniques, and setting a high merge effort for better compaction. You can refer to the TetraMax user manual posted on the Canvas project directory for more info on using the options.

For the first part of the project, you need to submit these files as one zip file. You are required to name each file exactly as listed below in red.

- (1) your .vg file generated by synthesis (**b14_fullscan.vg**)
- (2) your .stil file generated by synthesis for running DRC (**b14_fullscan.stil**)
- (3) your final test vectors written in binary format to allow external testing by us (**b14_pattern_fullscan.v**)
- (4) your script for full scan synthesis with design_vision (**synthesis_fullscan.tcl**)
- (5) your script for running TetraMax (**tmax_fullscan.tcl**)
- (6) one README file explaining how the two scripts should be used (e.g., directory to run each from) (**readme.txt**)
- (7) a one-page report containing (**lastname.pdf or .docx**)
 - a snapshot of your ATPG report (Uncollapsed Stuck Fault Summary Report and Pattern Summary Report)
 - a discussion on how ATPG was configured, for example how the options were set, to reach the reported test coverage and number of patterns

(2) Test Pattern Generation for Partial Scan:

You are required to develop a test set for the same design in the partial scan mode to maximize the metric below:

$$M = TC / (A/K_1 \times N/K_2 \times L/K_3 \times P/K_4)$$

with

TC: Test Coverage reported by TetraMax

A: Area after partial scan as reported by Design Vision

N: Number of patterns reported by TetraMax

L: Length of the maximum length chain as reported by Design Vision

P: Number of additional pins assuming each scan chain results in 3 extra pins

and

$$K_1 = 14233, K_2 = 480, K_3 = 215, K_4 = 3$$

The above parameters are defined based on the full scan characteristics.

In the above metric, L correlates with number of cycles to test one pattern so $L \times N$ estimates the test time on the ATE machine. Partial scan results in smaller area, fewer number of patterns, but lower test coverage. Higher value of M reflects a better overall design because all these factors should be considered simultaneously.

When designing partial scan to optimize M, assume that we cannot define more than 2 scan chains because of the pin count that the ATE machine can handle. **Your solution also needs to ensure a test coverage (TC) of at least 40%.**

For full scan mode, the above metric is very close to 100%. The goal is to show you can obtain a higher M than full scan mode using partial scan. Your project will be graded based on M calculated for your partial scan design.

There are two aspects to your partial scan design:

(1) Selecting the K most promising flipflops:

- Use your knowledge from the course to select a desired number (denoted by K) of the *most promising* flipflops for partial scan, e.g., using SCOAP testability measures. The second part of the project will also be individual work by default. However, you are allowed to work in a team of three if you pick flipflops for partial scan using the S-Graph technique; it requires additional software implementation for extracting a state transition graph for the design (second person), and to select flipflops to break the cycles in the S-Graph (third person).

(2) How many flipflops to select?

- Once you have a strategy to identify the top K most-promising flipflops, implement a design exploration technique to identify the K value that maximizes M. You are encouraged (not required) to develop automate using scripts.

For the second part of the project, you need to submit these files as **one** zip file. Name each file exactly as listed below in red. If you work in a group, please only submit one file per group.

- (1) your .vg file generated by synthesis (**b14_pscan.vg**)
- (2) your .stil file generated by synthesis for running DRC (**b14_pscan.stil**)
- (3) your final test vectors written in binary format to allow external testing by us (**b14_pattern_pscan.v**)
- (4) your script for partial scan synthesis with design_vision (**synthesis_pscan.tcl**)
- (5) your script for running TetraMax (**tmax_pscan.tcl**)
- (6) optional script(s) implementing design exploration to find value K
- (7) one README file listing the group member names, *and* explaining how the submitted scripts should be used (e.g., directory to run each from) (**readme.txt**)
- (8) a report (at most 2 pages) which should contain the following sections (**lastname(s).pdf or .docx**)

- Your name (or group member names)
- How the scan flipflops were selected for partial scan?
- How design exploration was conducted?
- Fill the table below which includes the design points that were explored by you to find your best design, highlighting the best column (highest M) which should correspond to your submitted partial scan files (items 1-3)

	Partial 1	Partial 2 (best)	...	Partial N
TC				
A				
N				
L				
P				
#Chains				
#Scan FFs				
M				

- If you worked in a group three, include these additional sections:
 - (1) how S-Graph was implemented, for example if you used certain report features of Design Vision to help with creating a netlist
 - (2) what strategies were implemented to select flipflops from the S-graph
 - (3) contributions of each team member: one possible break down of duties is that the first team member works on item 1 (S-graph extraction) and second and third team members work on item 2 (selection of flipflops from S-graph) and design space exploration